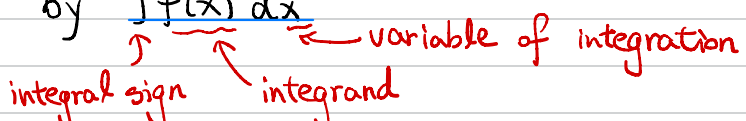


Antiderivatives & Indefinite Integrals

Antiderivatives: A function F is an antiderivative of f on an interval I if $F'(x) = f(x)$ for all x in I .

Let $F(x)$ be a function such that $F'(x) = f(x)$, then the most general antiderivative of f on I is $F(x) + C$, where C is an arbitrary constant.

Indefinite Integrals: The collection of all antiderivatives of f is called the indefinite integral of f with respect to x , and it is denoted by $\int f(x) dx$.



Examples: Let $f(x) = 2x$, and let C be an arbitrary constant. Which of the following statements is FALSE?

- A. $x^2 + 1$ is an antiderivative of $f(x)$
- B. x^2 is an antiderivative of $f(x)$
- ☒ C. x^2 is the indefinite integral of $f(x)$
- D. $x^2 + C$ is the indefinite integral of $f(x)$

(Riemann Sum)

Area and Estimating with Finite Sums

area below a positive & continuous function on $[a, b]$

Let $A = \text{area of } R$

Upper sum for A

$$= 1 \cdot \frac{1}{2} + \frac{3}{4} \cdot \frac{1}{2}$$

$$\text{or} \\ = 1 \cdot \frac{1}{4} + \frac{15}{16} \cdot \frac{1}{4} + \frac{3}{4} \cdot \frac{1}{4} + \frac{7}{16} \cdot \frac{1}{4}$$

Lower sum for A

$$= \frac{3}{4} \cdot \frac{1}{2} + 0 \cdot \frac{1}{2}$$

$$\text{or} \\ = \frac{15}{16} \cdot \frac{1}{4} + \frac{3}{4} \cdot \frac{1}{4} + \frac{7}{16} \cdot \frac{1}{4} + 0 \cdot \frac{1}{4}$$

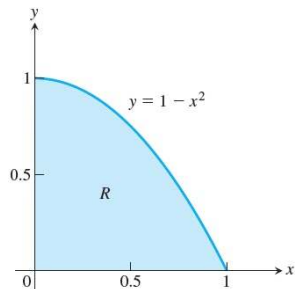


FIGURE 5.1 The area of a region R cannot be found by a simple formula.

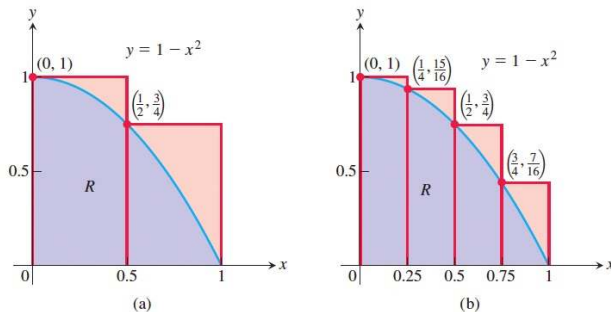


FIGURE 5.2 (a) We get an upper estimate of the area of R by using two rectangles containing R . (b) Four rectangles give a better upper estimate. Both estimates overshoot the true value for the area by the amount shaded in light red.

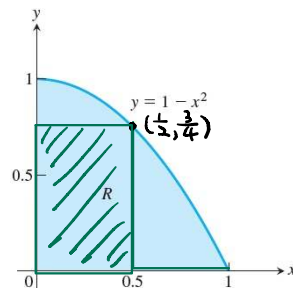


FIGURE 5.1 The area of a region R cannot be found by a simple formula.

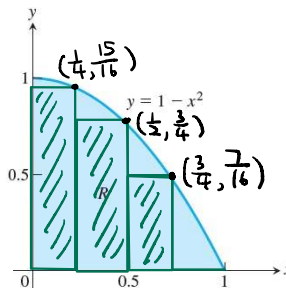


FIGURE 5.1 The area of a region R cannot be found by a simple formula.

Finite Sum

$$\text{e.g. } \sum_{i=1}^n \left(\frac{i^2}{n} + \frac{2i}{n^2} + 1 \right) = \frac{1}{n} \sum_{i=1}^n i^2 + \frac{2}{n^2} \sum_{i=1}^n i + \sum_{i=1}^n 1$$

$$\sum_{i=1}^n 1 = n$$

$$\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$$

$$\sum_{i=1}^n i = \frac{n(n+1)}{2}$$

$$\sum_{i=1}^n i^3 = \left(\frac{n(n+1)}{2} \right)^2$$

A Formula for the Riemann Sum with Equal Width Subintervals

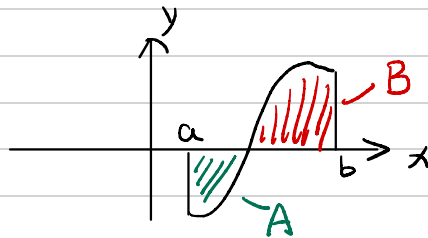
$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x = \lim_{n \rightarrow \infty} \sum_{i=1}^n f\left(a + i\left(\frac{b-a}{n}\right)\right) \left(\frac{b-a}{n}\right)$$

$$\text{average value} = \frac{1}{b-a} \int_a^b f(x) dx$$

Geometric Meaning of the Definite Integral

Net area below the graph of $f(x)$

$$\text{e.g. } \int_a^b f(x) dx = B - A$$



The Fundamental Theorem of Calculus (FTC)

Let f be a function that is continuous on the interval $[a, b]$, and let F be any antiderivative of f . Then:

$$\int_a^b f(x) dx = F(x) \Big|_a^b = F(b) - F(a)$$

Average value:
(if f is cts on $[a, b]$)

$$\frac{\text{Area}}{b-a} = \frac{\int_a^b f(x) dx}{b-a} = \frac{F(b) - F(a)}{b-a}$$

2nd FTC If f is cts on the interval $[a, b]$.

$$\frac{d}{dx} \left[\int_{a(x)}^{b(x)} f(t) dt \right] = f(b(x)) b'(x) - f(a(x)) a'(x)$$

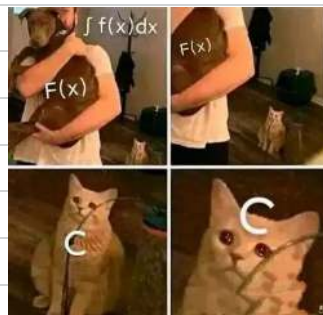
In particular $\frac{d}{dx} \left[\int_c^{b(x)} f(t) dt \right] = f(b(x)) b'(x)$

when c is a constant.

Integration by Substitution (u-sub)

Q: Which kind of integrands can the u-sub method be applied to?

A: When $f(x) f'(x)$ is included



Steps to Perform U-substitution

① Find $u=f(x)$

② Using dx to express du , i.e. $dx = \frac{1}{f'(x)} du$
b/c $\frac{du}{dx} = f'(x) \Rightarrow dx = \frac{1}{f'(x)} du$

③ Plugging ① & ② into the original question:

$$\int f(x) \cdot f'(x) dx = \int u \cdot \cancel{f'(x)} \left(\frac{1}{\cancel{f'(x)}} \right) du = \int u du$$

④ Evaluating & Plugging $u=f(x)$ back (Don't forget $+C$!)